1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights a tension—MMR defects (MSH6/MSI-dMMR) increase mutational burden and often T-cell infiltration, yet immune contexture (Immunoscore/TILs) does not always translate into effective anti-tumor immunity or predictable therapy benefit. Why does immune infiltration sometimes fail to produce clinical response in MSH6/MSI tumors?

Working Hypothesis

In bladder cancer with MSH6-driven MSI/dMMR, high TIL density indicates robust immune infiltration but does not universally predict checkpoint inhibitor (IO) therapy response. IO efficacy depends on (i) clonal neoantigen burden (favoring clonal over subclonal antigens) and (ii) intact antigen presentation (e.g., preserved HLA class I function, absence of B2M/HLA loss). Therefore, TIL density is a context-dependent predictor: it will correlate with therapy_response only in tumors with high clonal neoantigen load and preserved antigen presentation; in MSHand MSI-dMMR bladder tumors with subclonal neoantigens or impaired antigen presentation, high TIL density may not translate into clinical benefit. This rationale justifies combining TIL density with measures of neoantigen clonality and antigen presentation status to predict IO response, and incorporating noninvasive modifiers (urine DNA methylation, CTCs, AI-based imaging biomarkers) to refine patient selection. Testable predictions include: (a) among MSH6/MSI/dMMR bladder cancers treated with IO, responders will have higher clonal neoantigen burden and intact HLA presentation compared with non-responders, even when TIL density is high; (b) TIL-high tumors with low clonal neoantigen burden or evidence of HLA loss will show poorer therapy_response; (c) a composite model integrating TIL density, neoantigen clonality, and HLA status will outperform TIL density alone in predicting therapy_response. Feasible study design involves a bladder cancer cohort with known MSH6/MSI/dMMR status treated with IO, with data on TIL density, neoantigen clonality, HLA/B2M status, and noninvasive biomarkers, enabling multivariate modeling and validation in an independent set.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, MSI/dMMR
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6/MSI-dMMR) increase mutational burden and often T-cell infiltration, yet immune contexture (Immunoscore/TILs) does not always translate into effective anti-tumor immunity or predictable therapy benefit.

MSI/dMMR: MSI/dMMR because they directly explain the genomic source of immunogenicity while TIL density is the most direct imaging correlate of Immunoscore in routine pathology.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	urine_DNA_methylation_biomarkers	Supporting modifier from knowledge graph
Secondary	CTCs	Supporting modifier from knowledge graph
Secondary	AI_based_biomarker_accessibility	Supporting modifier from knowledge graph
Secondary	TP53/ER_status	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: antigen presentation integrity (HLA/B2M), and

Prediction 2: noninvasive modifiers (urine DNA methylation, CTCs, AI-derived biomarkers), we can derive a more accurate, testable predictor of IO therapy_response. The approach aligns with existing knowledge that, while mutational burden promotes immunogenicity, the quality and presentation of neoantigens determine durable clinical benefit.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- urine_DNA_methylation_biomarkers: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CTCs: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.